# DESIGN PATTERNS

## Exercise 1: Implementing the Singleton Pattern

### Scenario:

You need to ensure that a logging utility class in your application has only one instance throughout the application lifecycle to ensure consistent logging.

**CODE:**

```
class Logger {
    private static Logger instance;
    private Logger() {}
    public static Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
        return instance;
    }
    public void log(String message) {
        System.out.println("[LOG]: " + message);
    }
}
public class Main {
    public static void main(String[] args) {
        Logger logger1 = Logger.getInstance();
        Logger logger2 = Logger.getInstance();

        logger1.log("This is the first log message.");
        logger2.log("This is the second log message.");

        System.out.println("\n--- Verification Results ---");
        if (logger1 == logger2) {
            System.out.println("SUCCESS: Both variables point to the same instance!");
        } else {
            System.out.println("FAILURE: Different instances exist.");
        }
    }
}
```

Output

```
[LOG]: This is the first log message.
[LOG]: This is the second log message.

--- Verification Results ---
SUCCESS: Both variables point to the same instance!

=== Code Execution Successful ===
```

## Exercise 2: Implementing the Factory Method Pattern

**Scenario:**

You are developing a document management system that needs to create different types of documents (e.g., Word, PDF, Excel). Use the Factory Method Pattern to achieve this.

**CODE:**

```
interface Document {
   void open();
   void close();
}

class WordDocument implements Document {
   public void open() { System.out.println("Opening Word Document (.docx)..."); }
   public void close() { System.out.println("Closing Word Document."); }
}

class PdfDocument implements Document {
   public void open() { System.out.println("Opening PDF Document (.pdf)..."); }
   public void close() { System.out.println("Closing PDF Document."); }
}

class ExcelDocument implements Document {
   public void open() { System.out.println("Opening Excel Spreadsheet (.xlsx)..."); }
   public void close() { System.out.println("Closing Excel Spreadsheet."); }
}
abstract class DocumentFactory {
   public abstract Document createDocument();
}

class WordDocumentFactory extends DocumentFactory {
   public Document createDocument() { return new WordDocument(); }
}

class PdfDocumentFactory extends DocumentFactory {
   public Document createDocument() { return new PdfDocument(); }
}

class ExcelDocumentFactory extends DocumentFactory {
   public Document createDocument() { return new ExcelDocument(); }
}

public class Main {
   public static void main(String[] args) {
      System.out.println("--- Document Management System ---\n");

      DocumentFactory wordFactory = new WordDocumentFactory();
      Document wordDoc = wordFactory.createDocument();
      wordDoc.open();
      wordDoc.close();
```

```
        System.out.println();

        DocumentFactory pdfFactory = new PdfDocumentFactory();
        Document pdfDoc = pdfFactory.createDocument();
        pdfDoc.open();
        pdfDoc.close();

        System.out.println();

        DocumentFactory excelFactory = new ExcelDocumentFactory();
        Document excelDoc = excelFactory.createDocument();
        excelDoc.open();
        excelDoc.close();
    }
}
```

Output

```
--- Document Management System ---

Opening Word Document (.docx)...
Closing Word Document.

Opening PDF Document (.pdf)...
Closing PDF Document.

Opening Excel Spreadsheet (.xlsx)...
Closing Excel Spreadsheet.

=== Code Execution Successful ===
```

## Exercise 3: Implementing the Builder Pattern

### Scenario:

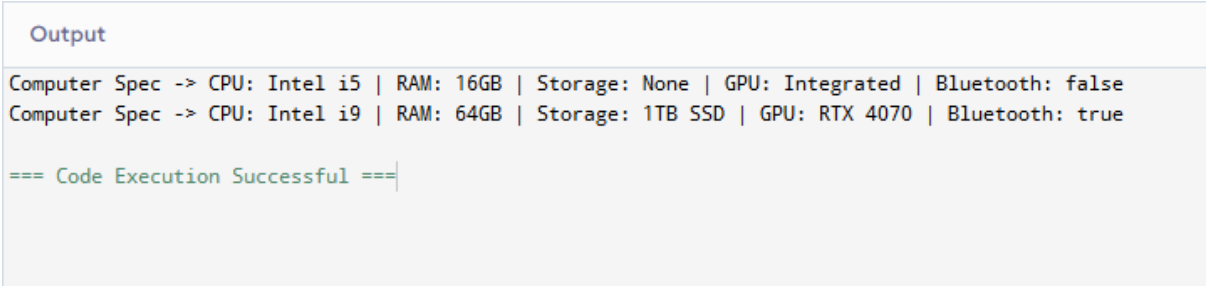
You are developing a system to create complex objects such as a Computer with multiple optional parts. Use the Builder Pattern to manage the construction process.

**CODE:**

```
class Computer {
    private final String CPU;
    private final String RAM;
    private final String storage;
    private final String graphicsCard;
    private final boolean isBluetoothEnabled;

    private Computer(Builder builder) {
        this.CPU = builder.CPU;
        this.RAM = builder.RAM;
        this.storage = builder.storage;
        this.graphicsCard = builder.graphicsCard;
        this.isBluetoothEnabled = builder.isBluetoothEnabled;
```

```
    }

    @Override
    public String toString() {
        return "Computer Spec -> CPU: " + CPU + " | RAM: " + RAM +
                " | Storage: " + (storage != null ? storage : "None") +
                " | GPU: " + (graphicsCard != null ? graphicsCard : "Integrated") +
                " | Bluetooth: " + isBluetoothEnabled;
    }

    public static class Builder {
        private final String CPU;
        private final String RAM;
        private String storage;
        private String graphicsCard;
        private boolean isBluetoothEnabled;

        public Builder(String CPU, String RAM) {
            this.CPU = CPU;
            this.RAM = RAM;
        }

        public Builder setStorage(String storage) {
            this.storage = storage;
            return this;
        }

        public Builder setGraphicsCard(String graphicsCard) {
            this.graphicsCard = graphicsCard;
            return this;
        }

        public Builder setBluetoothEnabled(boolean isBluetoothEnabled) {
            this.isBluetoothEnabled = isBluetoothEnabled;
            return this;
        }

        public Computer build() {
            return new Computer(this);
        }
    }
}

public class Main {
    public static void main(String[] args) {
        Computer standard = new Computer.Builder("Intel i5", "16GB").build();
        Computer premium = new Computer.Builder("Intel i9", "64GB")
                .setStorage("1TB SSD")
                .setGraphicsCard("RTX 4070")
                .setBluetoothEnabled(true)
                .build();

        System.out.println(standard);
        System.out.println(premium);
    }
```

}




## Exercise 4: Implementing the Adapter Pattern

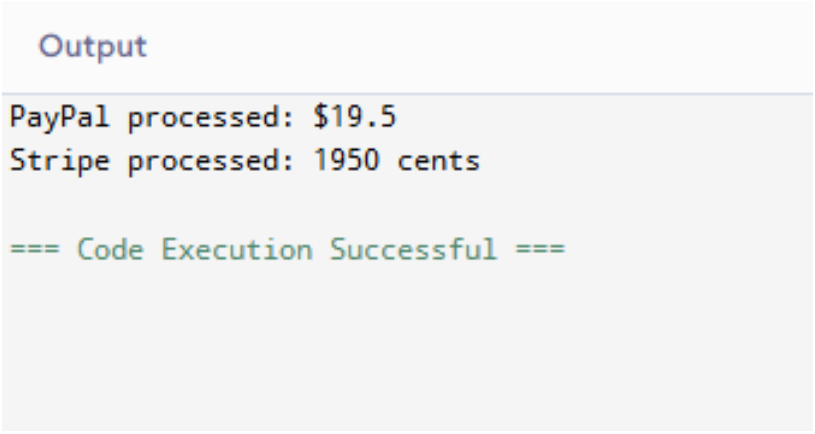
### Scenario:

You are developing a payment processing system that needs to integrate with multiple third-party payment gateways with different interfaces. Use the Adapter Pattern to achieve this.

### CODE:

```
interface PaymentProcessor {
    void processPayment(double amount);
}

class PayPalGateway {
    public void sendPayment(double totalUSD) {
        System.out.println("PayPal processed: $" + totalUSD);
    }
}

class StripeGateway {
    public void chargeCustomer(int amountInCents) {
        System.out.println("Stripe processed: " + amountInCents + " cents");
    }
}
class PayPalAdapter implements PaymentProcessor {
    private PayPalGateway gateway;
    public PayPalAdapter(PayPalGateway gateway) { this.gateway = gateway; }
    public void processPayment(double amount) { gateway.sendPayment(amount); }
}

class StripeAdapter implements PaymentProcessor {
    private StripeGateway gateway;
    public StripeAdapter(StripeGateway gateway) { this.gateway = gateway; }
    public void processPayment(double amount) { gateway.chargeCustomer((int)(amount * 100)); }
}

public class Main {
    public static void main(String[] args) {
        PaymentProcessor p1 = new PayPalAdapter(new PayPalGateway());
        PaymentProcessor p2 = new StripeAdapter(new StripeGateway());

        p1.processPayment(19.50);
        p2.processPayment(19.50);
    }
```

}




## Exercise 5: Implementing the Decorator Pattern

### Scenario:

You are developing a notification system where notifications can be sent via multiple channels (e.g., Email, SMS). Use the Decorator Pattern to add functionalities dynamically.

**CODE:**

```
interface Notifier {
    void send(String message);
}

class EmailNotifier implements Notifier {
    public void send(String message) {
        System.out.println("Email Out: " + message);
    }
}

abstract class NotifierDecorator implements Notifier {
    protected Notifier wrapper;
    public NotifierDecorator(Notifier notifier) { this.wrapper = notifier; }
    public void send(String message) { wrapper.send(message); }
}

class SMSNotifierDecorator extends NotifierDecorator {
    public SMSNotifierDecorator(Notifier notifier) { super(notifier); }
    public void send(String message) {
        super.send(message);
        System.out.println("SMS Out: " + message);
    }
}

class SlackNotifierDecorator extends NotifierDecorator {
    public SlackNotifierDecorator(Notifier notifier) { super(notifier); }
    public void send(String message) {
        super.send(message);
        System.out.println("Slack Out: " + message);
    }
```

```
}


public class Main {
    public static void main(String[] args) {
        System.out.println("Deploying Multi-Channel Notification Stack:");
        Notifier systemAlert = new SlackNotifierDecorator(new SMSNotifierDecorator(new
EmailNotifier()));
        systemAlert.send("System Update Successful.");
    }
}
```

```
Output

Deploying Multi-Channel Notification Stack:
Email Out: System Update Successful.
SMS Out: System Update Successful.
Slack Out: System Update Successful.

=== Code Execution Successful ===
```

## Exercise 6: Implementing the Proxy Pattern

### Scenario:

You are developing an image viewer application that loads images from a remote server. Use the Proxy Pattern to add lazy initialization and caching.

**CODE:**

```
interface Image {
    void display();
}

class RealImage implements Image {
    private String fileName;

    public RealImage(String fileName) {
        this.fileName = fileName;
        loadFromRemoteServer();
    }

    private void loadFromRemoteServer() {
        System.out.println("-> [Network Loading]: " + fileName);
    }
```

```
    public void display() {
        System.out.println("-> [Rendering Graphic]: " + fileName);
    }
}

class ProxyImage implements Image {
    private String fileName;
    private RealImage cacheInstance;

    public ProxyImage(String fileName) {
        this.fileName = fileName;
    }

    public void display() {
        if (cacheInstance == null) {
            cacheInstance = new RealImage(fileName);
        } else {
            System.out.println("-> [Cache Hit]: Serving matching asset from cache.");
        }
        cacheInstance.display();
    }
}

public class Main {
    public static void main(String[] args) {
        Image img = new ProxyImage("dashboard_analytics.png");

        System.out.println("--- Action 1 ---");
        img.display();

        System.out.println("\n--- Action 2 ---");
        img.display();
    }
}
```

Output

```
--- Action 1 ---
-> [Network Loading]: dashboard_analytics.png
-> [Rendering Graphic]: dashboard_analytics.png

--- Action 2 ---
-> [Cache Hit]: Serving matching asset from cache.
-> [Rendering Graphic]: dashboard_analytics.png

=== Code Execution Successful ===
```

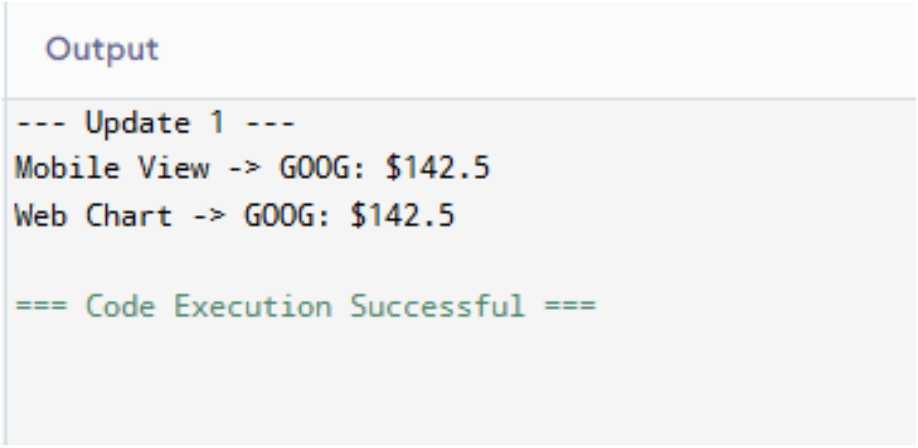
## Exercise 7: Implementing the Observer Pattern

### Scenario:

You are developing a stock market monitoring application where multiple clients need to be notified whenever stock prices change. Use the Observer Pattern to achieve this.

**CODE:**

```
import java.util.ArrayList;
import java.util.List;

interface Observer { void update(String stock, double price); }
interface Stock { void add(Observer o); void remove(Observer o); void notifyAllObservers(); }

class StockMarket implements Stock {
   private List<Observer> list = new ArrayList<>();
   private String sym; private double prc;

   public void add(Observer o) { list.add(o); }
   public void remove(Observer o) { list.remove(o); }
   public void notifyAllObservers() { for(Observer o : list) o.update(sym, prc); }

   public void updatePrice(String sym, double prc) {
      this.sym = sym; this.prc = prc;
      notifyAllObservers();
   }
}

class MobileApp implements Observer {
   public void update(String s, double p) { System.out.println("Mobile View -> " + s + ": $" + p); }
}
class WebApp implements Observer {
   public void update(String s, double p) { System.out.println("Web Chart -> " + s + ": $" + p); }
}

public class Main {
   public static void main(String[] args) {
      StockMarket market = new StockMarket();
      Observer m = new MobileApp();
      Observer w = new WebApp();

      market.add(m); market.add(w);
      System.out.println("--- Update 1 ---");
      market.updatePrice("GOOG", 142.50);
   }
}
```




## Exercise 8: Implementing the Strategy Pattern

**Scenario:**

You are developing a payment system where different payment methods (e.g., Credit Card, PayPal) can be selected at runtime. Use the Strategy Pattern to achieve this.

**CODE:**

```
interface PaymentStrategy {
    void pay(double amount);
}

class CreditCardPayment implements PaymentStrategy {
    private String name;
    public CreditCardPayment(String name) { this.name = name; }
    public void pay(double amount) {
        System.out.println("Paid $" + amount + " with Credit Card for owner " + name);
    }
}

class PayPalPayment implements PaymentStrategy {
    private String email;
    public PayPalPayment(String email) { this.email = email; }
    public void pay(double amount) {
        System.out.println("Paid $" + amount + " with PayPal account " + email);
    }
}

class PaymentContext {
    private PaymentStrategy strategy;
    public void setStrategy(PaymentStrategy strategy) { this.strategy = strategy; }
    public void executePayment(double amount) {
        strategy.pay(amount);
    }
}

public class Main {
```

```
    public static void main(String[] args) {
        PaymentContext context = new PaymentContext();

        System.out.println("--- Action 1 ---");
        context.setStrategy(new CreditCardPayment("Alice"));
        context.executePayment(25.00);

        System.out.println("\n--- Action 2 ---");
        context.setStrategy(new PayPalPayment("alice@example.com"));
        context.executePayment(42.50);
    }
}
```

Output

```
--- Action 1 ---
Paid $25.0 with Credit Card for owner Alice

--- Action 2 ---
Paid $42.5 with PayPal account alice@example.com

=== Code Execution Successful ===
```

## Exercise 9: Implementing the Command Pattern

**Scenario:** You are developing a home automation system where commands can be issued to turn devices on or off. Use the Command Pattern to achieve this.

**CODE:**

```
class Light {
    public void turnOn() { System.out.println("Light state: ON"); }
    public void turnOff() { System.out.println("Light state: OFF"); }
}

interface Command { void execute(); }

class LightOnCommand implements Command {
    private Light light;
    public LightOnCommand(Light light) { this.light = light; }
    public void execute() { light.turnOn(); }
}

class LightOffCommand implements Command {
    private Light light;
    public LightOffCommand(Light light) { this.light = light; }
    public void execute() { light.turnOff(); }
```

```
}

class RemoteControl {
    private Command cmd;
    public void setCommand(Command cmd) { this.cmd = cmd; }
    public void pressButton() { cmd.execute(); }
}

public class Main {
    public static void main(String[] args) {
        Light homeLight = new Light();
        RemoteControl remote = new RemoteControl();

        remote.setCommand(new LightOnCommand(homeLight));
        remote.pressButton();

        remote.setCommand(new LightOffCommand(homeLight));
        remote.pressButton();
    }
}
```

Output

```
Light state: ON
Light state: OFF

=== Code Execution Successful ===
```

## Exercise 10: Implementing the MVC Pattern

**Scenario:**

You are developing a simple web application for managing student records using the MVC pattern.

**CODE:**

```
class Student {
    private String name, id, grade;
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getId() { return id; }
    public void setId(String id) { this.id = id; }
    public String getGrade() { return grade; }
    public void setGrade(String grade) { this.grade = grade; }
```

```
}

class StudentView {
    public void displayStudentDetails(String n, String i, String g) {
        System.out.println("Profile -> ID: " + i + " | Name: " + n + " | Grade: " + g);
    }
}

class StudentController {
    private Student model;
    private StudentView view;
    public StudentController(Student m, StudentView v) { this.model = m; this.view = v; }
    public void setStudentName(String name) { model.setName(name); }
    public void setStudentGrade(String grade) { model.setGrade(grade); }
    public void updateView() { view.displayStudentDetails(model.getName(), model.getId(),
model.getGrade()); }
}

public class Main {
    public static void main(String[] args) {
        Student s = new Student();
        s.setName("Alice"); s.setId("101"); s.setGrade("B");

        StudentController ctrl = new StudentController(s, new StudentView());
        ctrl.updateView();

        ctrl.setStudentName("Bob");
        ctrl.setStudentGrade("A");
        ctrl.updateView();
    }
}
```

Output

```
Profile -> ID: 101 | Name: Alice | Grade: B
Profile -> ID: 101 | Name: Bob | Grade: A

=== Code Execution Successful ===
```

## Exercise 11: Implementing Dependency Injection

### Scenario:

You are developing a customer management application where the service class depends on a repository class. Use Dependency Injection to manage these dependencies.

### CODE:

```
interface CustomerRepository {
    String findCustomerById(int id);
}

class CustomerRepositoryImpl implements CustomerRepository {
    public String findCustomerById(int id) {
        return (id == 7) ? "John Doe" : "Record Not Found";
    }
}

class CustomerService {
    private final CustomerRepository repository;

    public CustomerService(CustomerRepository repository) {
        this.repository = repository;
    }

    public void getCustomerName(int id) {
        System.out.println("Resolved Name: " + repository.findCustomerById(id));
    }
}

public class Main {
    public static void main(String[] args) {

        CustomerRepository repo = new CustomerRepositoryImpl();
        CustomerService service = new CustomerService(repo);

        service.getCustomerName(7);
        service.getCustomerName(12);
    }
}
```

Output

```
Resolved Name: John Doe
Resolved Name: Record Not Found

=== Code Execution Successful ===
```